

Technical Document #48: Retrieval Augmented Generation - Comprehensive Overview

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Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Semantic search finds documents based on meaning rather than exact keywords.
- Hallucination detection identifies when LLMs generate factually incorrect information.